EVERY REMAINDER EVERYWHERE

SIO

Abstract. We prove that for every prime p, every positive integer w not divisible by 3, every residue class $r \in \mathbb{Z}/p\mathbb{Z}$, and every threshold $B \in \mathbb{N}$, there exists an integer $n > B$ such that n reaches w under iterations of the Collatz map and $n \equiv r \pmod{p}$. In particular, the set of reaching predecessors of w in any residue class modulo p is infinite. The proof proceeds by analyzing an explicit two-branch affine dynamical system on the 3-adic integers. We define a family of finite fibres recording the residues modulo p attained by descendants of an adic seed; show that their cardinalities stabilize; and prove that an affine holonomy group acts transitively on the stabilized fibres via a non-trivial translation generated by a group commutator. The entire mathematical development is formalization-faithful, matching a complete verification in Lean 4. Finally, as an application, we establish that single-prime modular obstructions cannot rule out hypothetical Collatz cycle equations.

Standing Assumptions and Conventions

Throughout this paper, p denotes a fixed prime number. In Sections 6 through 8, we assume $p \geq 5$. The ring $\mathbb{Z}/p\mathbb{Z}$ denotes the finite field of order p, and $(n : \mathbb{Z}/p\mathbb{Z})$ denotes the canonical image of $n \in \mathbb{N}$ in $\mathbb{Z}/p\mathbb{Z}$. All divisions between natural numbers defining our operators are exact in $\mathbb{N}$ (as established in Lemma 4.1), mirroring the verified formalization in Lean 4. Explicit truncating division $\lfloor \cdot \rfloor$ is used only in Proposition 5.7.

1. Collatz Dynamics and the Main Result

The standard $3n + 1$ Collatz operator $\mathcal{C} \colon \mathbb{N} \to \mathbb{N}$ is defined by:

$$\mathcal{C}(n) = \begin{cases} n/2 & \text{if } n \equiv 0 \pmod 2, \\ 3n+1 & \text{if } n \equiv 1 \pmod 2. \end{cases}$$

We say n *reaches* v (written $n \rightsquigarrow v$) if $\mathcal{C}^{(k)}(n) = v$ for some $k \geq 0$.

Theorem 1.1 (Main Theorem)**.** *Let p be a prime. Let $w \in \mathbb{N}$ with $w > 0$ and $3 \nmid w$, let $r \in \mathbb{Z}/p\mathbb{Z}$, and let $B \in \mathbb{N}$. Then there exists $n \in \mathbb{N}$ such that $n > B$, $n \rightsquigarrow w$, and $(n : \mathbb{Z}/p\mathbb{Z}) = r$.*

The proof of Theorem 1.1 splits into two regimes:

- **Primes $p \geq 5$:** We construct an abstract descendant tree $\mathcal{D}(v)$ from a seed $v \in G$ reaching w, and establish a density theorem (Theorem 8.6): in every valid cylinder at depth K, descendants of v realize *every* residue modulo p. Choosing K large enough ensures $n > B$.
- **Small primes $p \in \{2, 3\}$:** Explicit backward paths under pure halvings and division by 3 resolve all cases directly (Section 9).

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Lemma 1.2 (Halving and Atomic Inverse Steps)**.**
(1) $\mathcal{C}(2a) = a$*, and* $\mathcal{C}^{(j)}(2^j m) = m$ *for all* $j, m \in \mathbb{N}$*.*
(2) Reachability is transitive: if $a \rightsquigarrow b$ *and* $b \rightsquigarrow c$*, then* $a \rightsquigarrow c$*.*
(3) For all $m, n \in \mathbb{N}$*:* $\mathcal{C}(m) = n$ *if and only if* $m = 2n$*, or* m *is odd and* $3m + 1 = n$*.*

Proof. (1) and (2) are immediate. (3) is the definition of $\mathcal{C}$ read backwards: if m is even then $\mathcal{C}(m) = m/2$, so $\mathcal{C}(m) = n$ means $m = 2n$; if m is odd then $\mathcal{C}(m) = 3m + 1$. □

Thus a backward step from n has at most two options: *double* ($n \mapsto 2n$, always available), or *undo* $3m + 1$ (available only when $n = 3m + 1$ with m odd). Everything that follows is a bookkeeping of compositions of these two atomic moves.

Corollary 1.3 (Multiples of 3 Admit Only Doubling)**.** *If* $3 \mid n$ *and* $\mathcal{C}(m) = n$*, then* $m = 2n$*.*

Proof. If m is odd with $3m + 1 = n$, then $n \equiv 1 \pmod 3$, contradicting $3 \mid n$. By Lemma 1.2(3), $m = 2n$. □

The corollary is the reason the hypothesis $3 \nmid w$ appears in Theorem 1.1, and the reason only odd numbers prime to 3 can branch.

2. Good Numbers and the Coordinate $x = (3n + 1)/2$

Definition 2.1 (Good Numbers and the Coordinate)**.** An integer $n \in \mathbb{N}$ is *good* if n is odd and $3 \nmid n$. For odd n put

$$x(n) = \frac{3n + 1}{2}.$$

Lemma 2.2 (Coordinate Step)**.** *If* n *is odd, then* $\mathcal{C}(n) = 2x(n)$ *and* $\mathcal{C}(2x(n)) = x(n)$*. In particular* $n \rightsquigarrow x(n)$ *in two steps.*

Proof. n odd gives $\mathcal{C}(n) = 3n + 1 = 2x(n)$, and then $\mathcal{C}(2x(n)) = x(n)$ by Lemma 1.2(1). □

The coordinate x fuses the two forced steps $n \mapsto 3n + 1 \mapsto (3n + 1)/2$ into one. Only the odd, 3-free numbers are branch points, and the next lemma identifies them in x-coordinates.

Definition 2.3 (Cylinders and Valid Classes)**.** A natural number $c \in \mathbb{N}$ is called *valid* if $c \equiv 2 \pmod 9$ or $c \equiv 8 \pmod 9$. A *valid class at depth* K (for $K \geq 1$) is a valid integer satisfying the boundedness condition $c < 3^{K+1}$. Modulo 3^{K+1}, the valid classes represent the admissible 3-adic residue cylinders of our inverse operators.

Definition 2.4 (Seed Set)**.** The admissible seed set is

$$G = \{x \in \mathbb{N} : x \geq 8 \text{ and } (x \equiv 2 \pmod 9 \ \vee \ x \equiv 8 \pmod 9)\}.$$

Lemma 2.5 (Good Numbers Are the Valid Classes)**.** *If* $n \geq 5$ *is good, then* $x(n) \geq 8$ *and* $x(n)$ *is valid. Conversely, if* $x \geq 8$ *is valid, then* $n = (2x - 1)/3$ *is a good integer with* $n \geq 5$ *and* $x(n) = x$*. Hence*

$$G = \{\, x(n) : n \geq 5 \text{ good} \}.$$

Proof. For odd n we have $x(n)$ mod 9 determined by n mod 18. The odd residues prime to 3 modulo 18 are $1, 5, 7, 11, 13, 17$, and they give $x(n) \equiv 2, 8, 2, 8, 2, 8 \pmod 9$ respectively. The odd residues $3, 9, 15$ (multiples of 3) give $x(n) \equiv 5, 5, 5 \pmod 9$ and are excluded. The bound $n \geq 5$ gives $x(n) \geq 8$; the only good number below 5 is $n = 1$, whose coordinate $x(1) = 2$ lies on the trivial cycle $1 \to 4 \to 2 \to 1$ and is excluded by the threshold $x \geq 8$. Conversely, if $x \equiv 2$ or $8 \pmod 9$ then $2x - 1 \equiv 3$ or $15 \pmod{18}$, so $n = (2x - 1)/3$ is an integer, odd (as $3n = 2x - 1$ is odd), $n \equiv 1$ or $5 \pmod 6$, hence good; $x \geq 8$ gives $n \geq 5$, and $x(n) = x$ by direct computation. □

Remark 2.6 (On the Projection)**.** The coordinate x is a bookkeeping choice, not an extra hypothesis: Lemma 2.2 shows each original-coordinate node n reaches its x-coordinate in two Collatz steps. Its benefit is that the right moves become pure multiplications ($4x$ or $16x$), so the descendant tree is a clean binary tree on a single set of residue classes. Proposition 3.3(3) below shows that the original-coordinate rules ($4n+1$, $16n+5$, and the mod-9 left rule) are carried to R and L exactly.

3. The Branch Operators as Composite Inverse Steps

Definition 3.1 (Branch Operators and Multipliers)**.** For $x \in \mathbb{N}$, define the right and left inverse branches:

$$R(x) = \begin{cases} 4x & \text{if } x \equiv 2 \pmod 9, \\ 16x & \text{if } x \not\equiv 2 \pmod 9, \end{cases} \qquad \text{and} \qquad L(x) = \frac{\mu(x)(2x-1)}{6},$$

where the multiplier $\mu(x) \in \{2, 4, 8, 16\}$ is determined by $x \bmod 27$:

$$\mu(x) = \begin{cases} 4 & \text{if } x \equiv 2 \text{ or } 20 \pmod{27}, \\ 8 & \text{if } x \equiv 8 \pmod{27}, \\ 16 & \text{if } x \equiv 11 \pmod{27}, \\ 2 & \text{if } x \equiv 17 \text{ or } 26 \pmod{27}. \end{cases}$$

Definition 3.2 (Abstract Descendant Tree)**.** For any $v \in G$, the descendant set $\mathcal{D}(v)$ is the smallest subset of $\mathbb{N}$ containing v and closed under R and L.

Proposition 3.3 (Derivation of the Branches)**.** *Let $x \in G$.*

(1) (Right branch.) *$R(x)$ is the first valid element of the doubling chain $2x, 4x, 8x, 16x$. Explicitly, if $x \equiv 2 \pmod 9$ then $2x$ is not valid and $R(x) = 4x$; if $x \equiv 8 \pmod 9$ then $2x, 4x, 8x$ are not valid and $R(x) = 16x$.*

(2) (Left branch.) *Let $m = (2x-1)/3$. Then m is odd, $\mathcal{C}(m) = 2x$ (the backward $3m+1$ step), and $L(x)$ is the first valid element of $m, 2m, 4m, 8m$.*

(3) (Original coordinates.) *Let n be good and define*

$$R_0(n) = \begin{cases} 4n+1 & n \equiv 1 \pmod 3, \\ 16n+5 & n \equiv 2 \pmod 3, \end{cases} \qquad L_0(n) = \frac{\mu_0(n)\, n - 1}{3},$$

where $\mu_0(n) = 4, 2, 4, 8, 16, 2$ for $n \equiv 1, 2, 4, 5, 7, 8 \pmod 9$. Then $R(x(n)) = x(R_0(n))$ and $L(x(n)) = x(L_0(n))$.

Proof. (1) If $x \equiv 2 \pmod 9$, then $2x, 4x$ have residues $4, 8$ modulo 9, so the first valid one is $4x$. If $x \equiv 8 \pmod 9$, then $2x, 4x, 8x, 16x$ have residues $7, 5, 1, 2$, so the first valid one is $16x$.

(2) Since $3m = 2x - 1$ is odd, m is odd, and $\mathcal{C}(m) = 3m + 1 = 2x$ by Lemma 1.2(3). Further $L(x) = \frac{\mu(x)}{2}\, m$, by Lemma 4.1 below, with $\mu(x)/2 \in \{1, 2, 4, 8\}$. Running through the six classes of $m \bmod 9$ (namely $1, 2, 4, 5, 7, 8$, with $m = n$ in the notation of Lemma 2.5) the first of $m, 2m, 4m, 8m$ that is $\equiv 2, 8 \pmod 9$ is $2m, m, 2m, 4m, 8m, m$ respectively, matching $\mu(x)/2 = 2, 1, 2, 4, 8, 1$.

(3) For good n, $x = x(n)$ satisfies $2x - 1 = 3n$. We have $x \equiv 2 \pmod 9$ iff $3n + 1 \equiv 4 \pmod{18}$ iff $n \equiv 1 \pmod 3$. In that case $x(4n+1) = 6n + 2 = 4x(n)$; otherwise $x(16n+5) = 24n + 8 = 16x(n)$. For L: the residues $n \equiv 1, 5, 7, 11, 13, 17 \pmod{18}$ give $x \equiv 2, 8, 11, 17, 20, 26 \pmod{27}$, so $\mu(x) = 4, 8, 16, 2, 4, 2$, which is exactly $\mu_0(n)$ for $n \bmod 9 = 1, 5, 7, 2, 4, 8$. Hence $L(x) = \frac{\mu(x)}{2}\, n = \frac{\mu_0(n)}{2}\, n$, while $x(L_0(n)) = (3L_0(n) + 1)/2 = \frac{\mu_0(n)}{2}\, n$ (recall $\mu_0(n)$ is even). $\square$

Proposition 3.3 is the precise sense in which the tree is *derived* from the Collatz map rather than posited: every branch is a composite of the two atomic moves of Lemma 1.2(3), cut off at the first valid class.

Lemma 3.4 (Branch Trajectories)**.**

(1) For all $x \in \mathbb{N}$, $R(x) \rightsquigarrow x$. Specifically, $R(x)$ reaches x in 2 steps if $x \equiv 2 \pmod 9$, and in 4 steps otherwise.

(2) If $x \geq 1$ and $x \equiv 2 \pmod 3$, then $L(x) \rightsquigarrow x$.

Proof. (1) follows from $R(x) = 2^2x$ or 2^4x and Lemma 1.2(1). For (2), by Lemma 4.1 below, $6L(x) = \mu(x)(2x-1)$ holds identically. Since $\mu(x) \in \{2, 4, 8, 16\}$, we write $\mu(x) = 2 \cdot 2^i$ with $i \in \{0, 1, 2, 3\}$. Since $x \equiv 2 \pmod 3$, $2x - 1 \equiv 3 \equiv 0 \pmod 3$, so $m = (2x-1)/3$ is an integer. Thus $6L(x) = 6 \cdot 2^i m$, which gives $L(x) = 2^i m$. Because $3m = 2x - 1$ is odd, m is necessarily odd. Hence $\mathcal{C}(m) = 3m + 1 = 2x$, and $\mathcal{C}(2x) = x$. Therefore, $L(x) = 2^i m$ reaches x in $i + 2$ iterations of $\mathcal{C}$. □

Lemma 3.5 (Soundness of the Tree)**.** *If $v \in G$ and $n \in \mathcal{D}(v)$, then $n \rightsquigarrow v$.*

Proof. We argue by induction on the derivation of $n \in \mathcal{D}(v)$. The base case $n = v$ requires 0 steps. If $n = R(m)$ with $m \in \mathcal{D}(v)$, Lemma 3.4(1) gives $R(m) \rightsquigarrow m$, and $m \rightsquigarrow v$ by the induction hypothesis; reachability is transitive. If $n = L(m)$ with $m \in \mathcal{D}(v)$, Lemma 4.3 below shows $m \in G$, whence $m \geq 8 \geq 1$ and $m \equiv 2 \pmod 3$. By Lemma 3.4(2), $L(m) \rightsquigarrow m$, and transitivity yields $L(m) \rightsquigarrow v$. □

Lemma 3.6 (Existence of Admissible Seeds)**.** *If $w > 0$ and $3 \nmid w$, there exists $v \in G$ such that $v \rightsquigarrow w$.*

Proof. Since $3 \nmid w$, $w \bmod 9 \in \{1, 2, 4, 5, 7, 8\}$. We select $v = 2^k w$ by choosing k according to the following assignment:

$w \bmod 9$	1	2	4	5	7	8
v	$128w$	$64w$	$128w$	$256w$	$512w$	$64w$
k	7	6	7	8	9	6
$v \bmod 9$	2	2	8	2	2	8

In each case, $v \geq 64 \geq 8$, $v \equiv 2$ or $8 \pmod 9$, so $v \in G$, and $\mathcal{C}^{(k)}(v) = w$ by Lemma 1.2(1). □

4. Branch Arithmetic and 3-Adic Precision Loss

Lemma 4.1 (Exactness of Left Division)**.** *If $x \equiv 2 \pmod 3$, then $6L(x) = \mu(x)(2x-1)$ holds in $\mathbb{N}$.*

Proof. By definition, $\mu(x) \in \{2, 4, 8, 16\}$, so $2 \mid \mu(x)$. Since $x \equiv 2 \pmod 3$, $2x - 1 \equiv 2(2) - 1 = 3 \equiv 0 \pmod 3$. Thus $2 \mid \mu(x)$ and $3 \mid (2x-1)$, which implies $6 \mid \mu(x)(2x-1)$. The division defining $L(x)$ is therefore exact. □

Lemma 4.2 (Residue Classification Modulo 27)**.** *If $x \equiv 2 \pmod 9$ or $x \equiv 8 \pmod 9$, then*

$$x \bmod 27 \in \{2, 8, 11, 17, 20, 26\}.$$

Proof. These six residue classes represent the complete preimage of $\{2, 8\}$ under the projection $\mathbb{Z}/27\mathbb{Z} \to \mathbb{Z}/9\mathbb{Z}$. □

Lemma 4.3 (Invariance of the Domain)**.** *Let $x \in G$.*

(1) $R(x) \in G$.

(2) $L(x) \in G$, and $L(x)$ is valid even without the threshold hypothesis $x \geq 8$.

(3) For any $v \in G$ and $n \in \mathcal{D}(v)$, $n \in G$.

Proof. (1) If $x \equiv 2 \pmod 9$, then $R(x) = 4x \equiv 8 \pmod 9$ and $R(x) \geq 4(8) = 32 \geq 8$. If $x \equiv 8 \pmod 9$, then $R(x) = 16x \equiv 7 \cdot 8 = 56 \equiv 2 \pmod 9$ and $R(x) \geq 16(8) \geq 8$.

(2) Since $x \in G$, $x \equiv 2 \pmod 3$, so Lemma 4.1 applies. Since $x \equiv a \pmod{27}$ with $a \in \{2, 8, 11, 17, 20, 26\}$, $L(x) \bmod 9$ depends only on $x \bmod 27$. Evaluating $L(a)$ on the representative $a \in \mathbb{N}$ and reducing modulo 9:

- $x \equiv 2 \implies L(2) = 4(3)/6 = 2 \equiv 2$; $\quad x \equiv 8 \implies L(8) = 8(15)/6 = 20 \equiv 2 \pmod 9$;
- $x \equiv 11 \implies L(11) = 16(21)/6 = 56 \equiv 2$; $\quad x \equiv 17 \implies L(17) = 2(33)/6 = 11 \equiv 2 \pmod 9$;
- $x \equiv 20 \implies L(20) = 4(39)/6 = 26 \equiv 8$; $\quad x \equiv 26 \implies L(26) = 2(51)/6 = 17 \equiv 8 \pmod 9$.

In all cases, $L(x) \equiv 2$ or $8 \pmod 9$. Moreover, for $x \geq 8$, the minimum value is $L(8) = 8(15)/6 = 20 \geq 8$.

(3) Follows by structural induction on $n \in \mathcal{D}(v)$ using (1) and (2). □

Lemma 4.4 (Precision Loss under L)**.** *Let $K \geq 1$. If $x \equiv y \pmod{3^{K+2}}$ with $x \equiv y \equiv 2 \pmod 3$, then $\mu(x) = \mu(y)$ and*

$$L(x) \equiv L(y) \pmod{3^{K+1}}.$$

Proof. Because $K \geq 1$, $3^3 \mid 3^{K+2}$, so $x \equiv y \pmod{27}$. By Definition 3.1, $\mu(x) = \mu(y)$. Over $\mathbb{Z}$, Lemma 4.1 gives:

$$6(L(x) - L(y)) = \mu(x)(2x - 1) - \mu(x)(2y - 1) = 2\,\mu(x)(x - y).$$

Dividing by 2 in $\mathbb{Z}$ yields $3(L(x) - L(y)) = \mu(x)(x - y)$. Since $3^{K+2} \mid (x - y)$, there exists $q \in \mathbb{Z}$ such that $x - y = 3^{K+2}q$. Substituting and dividing by 3 gives:

$$L(x) - L(y) = 3^{K+1}\,\mu(x)q.$$

Thus $L(x) \equiv L(y) \pmod{3^{K+1}}$. □

5. Right-Branch Dynamics on Cylinders

Fix $K \geq 1$. Modulo 3^{K+1}, the right branch acts as a discrete dynamical system on valid classes. Note that the results in this section are purely 3-adic and independent of p.

Definition 5.1 (Cylinder Maps and Step Exponents)**.** For a valid class $c < 3^{K+1}$, define $\rho(c) = R(c) \bmod 3^{K+1}$, and for a valid class $e < 3^{K+2}$, define $\lambda(e) = L(e) \bmod 3^{K+1}$. Define the right-step exponent:

$$\text{step}(c) = \begin{cases} 1 & \text{if } c \equiv 2 \pmod 9, \\ 2 & \text{if } c \not\equiv 2 \pmod 9, \end{cases}$$

so that $R(c) = c \cdot 4^{\text{step}(c)}$. The n-step iterate $\rho^n(c)$ and the cumulative exponent $E(n, c)$ are defined recursively by:

$$\rho^0(c) = c, \qquad \rho^{n+1}(c) = \rho^n(\rho(c)),$$
$$E(0, c) = 0, \qquad E(n + 1, c) = \text{step}(c) + E(n, \rho(c)).$$

Lemma 5.2. *Let $c < 3^{K+1}$ be a valid class. For all $n \in \mathbb{N}$:*

$$\rho^n(c) = \left(c \cdot 4^{E(n,c)}\right) \bmod 3^{K+1}.$$

Proof. By induction on n. The base case $n = 0$ is trivial. For the step, $R(c) = c \cdot 4^{\mathrm{step}(c)}$ and $\rho(c) = R(c) \bmod 3^{K+1}$. Since $\rho(c)$ is valid and bounded, the induction hypothesis gives $\rho^n(\rho(c)) \equiv \rho(c) \cdot 4^{E(n,\rho(c))} \equiv c \cdot 4^{\mathrm{step}(c)+E(n,\rho(c))} \pmod{3^{K+1}}$. Because $\rho^{n+1}(c) < 3^{K+1}$, the remainder coincides with the value. □

Lemma 5.3 (Two-Step Tripling)**.** *Let c be valid. Then* $\mathrm{step}(c) + \mathrm{step}(\rho(c)) = 3$. *Consequently, for any valid class $c < 3^{K+1}$ and any $k \in \mathbb{N}$:*

$$E(2k, c) = 3k.$$

Proof. R alternates the residue modulo 9: if $c \equiv 2$, then $R(c) \equiv 8$, so $\mathrm{step}(c) + \mathrm{step}(\rho(c)) = 1 + 2 = 3$; if $c \equiv 8$, then $R(c) \equiv 2$, so $2 + 1 = 3$. The second statement follows by induction on k. □

Lemma 5.4 (Lifting the Exponent for Base 4 at Prime 3)**.** *For any $n \geq 1$, $v_3(4^n - 1) = 1 + v_3(n)$. Consequently, for all $K, m \in \mathbb{N}$:*

$$4^m \equiv 1 \pmod{3^{K+1}} \iff 3^K \mid m.$$

Proof. The valuation identity is the classical LTE lemma for the base $(4, 1)$ at prime 3: $v_3(4^n - 1) = v_3(4 - 1) + v_3(n) = 1 + v_3(n)$. For the equivalence: if $m = 0$, both sides hold trivially. For $m \geq 1$, $4^m \equiv 1 \pmod{3^{K+1}} \iff v_3(4^m - 1) \geq K + 1 \iff 1 + v_3(m) \geq K + 1 \iff 3^K \mid m$. □

Lemma 5.5 (Four-Cancellation)**.** *If $a \leq b$ and $4^a \equiv 4^b \pmod{3^{K+1}}$, then $a \equiv b \pmod{3^K}$.*

Proof. Since $\gcd(4^a, 3^{K+1}) = 1$, we divide by 4^a to obtain $4^{b-a} \equiv 1 \pmod{3^{K+1}}$. Lemma 5.4 implies $3^K \mid (b - a)$. □

Lemma 5.6 (Exponents via Pigeonhole)**.** *For every valid class $c < 3^{K+1}$, there exists a unique $E \in \{0, 1, \ldots, 3^K - 1\}$ such that*

$$c = \left(2 \cdot 4^E\right) \bmod 3^{K+1}.$$

Moreover, $c \equiv 2 \pmod 9 \implies E \equiv 0 \pmod 3$, and $c \equiv 8 \pmod 9 \implies E \equiv 1 \pmod 3$.

Proof. Consider the target set $W = \{w < 3^{K+1} : w \equiv 2 \pmod 3\}$. Every $w \in W$ can be uniquely written as $3j + 2$ for $0 \leq j < 3^K$, so $|W| = 3^K$. Define $\psi\colon \{0, \ldots, 3^K - 1\} \to W$ by $\psi(E) = (2 \cdot 4^E) \bmod 3^{K+1}$. Since $4 \equiv 1 \pmod 3$, $2 \cdot 4^E \equiv 2 \pmod 3$, so $\psi(E) \in W$.

If $\psi(E_1) = \psi(E_2)$, then $2 \cdot 4^{E_1} \equiv 2 \cdot 4^{E_2} \pmod{3^{K+1}}$. Since $\gcd(2, 3^{K+1}) = 1$, $4^{E_1} \equiv 4^{E_2} \pmod{3^{K+1}}$. By Lemma 5.5, $E_1 \equiv E_2 \pmod{3^K}$. As both are in the range, $E_1 = E_2$. Being an injective map between finite sets of cardinality 3^K, ψ is surjective. Since valid classes have $c \equiv 2 \pmod 3$, $c \in W$, so $c = \psi(E)$ for some E.

For the residue condition: $4^E \bmod 9$ is $1, 4, 7$ according as $E \bmod 3$ is $0, 1, 2$. Therefore, $2 \cdot 4^E \bmod 9$ is $2, 8, 5$. Since c is valid, the residue cannot be 5, forcing the claimed classification. □

Proposition 5.7 (Transitivity of ρ)**.** *For any two valid classes $c, d < 3^{K+1}$, there exists $n \in \mathbb{N}$ such that $\rho^n(c) = d$.*

Proof. By Lemma 5.6, write $c \equiv 2 \cdot 4^{E_c}$ and $d \equiv 2 \cdot 4^{E_d} \pmod{3^{K+1}}$. Let $q = 3^{K-1} E_c$ (noting $K \geq 1$). Set $y = E_d + 3 \cdot q - E_c = E_d + 3^K E_c - E_c \geq 0$.

We claim there exists n with $E(n, c) = y$. If $y \equiv 0 \pmod 3$, choose $n = 2(y/3)$; by Lemma 5.3, $E(n, c) = 3(y/3) = y$. If $y \equiv 1$ or $2 \pmod 3$, take $n = 2\lfloor y/3 \rfloor + 1$, so $E(n, c) = \mathrm{step}(c) + 3\lfloor y/3 \rfloor$. Reducing $y + E_c = E_d + 3^K E_c$ modulo 3 gives $y \equiv E_d - E_c$, where $E_c, E_d \in \{0, 1\} \pmod 3$ by

Lemma 5.6. If $y \equiv 1$, then $E_c \equiv 0$ and $E_d \equiv 1$, so $c \equiv 2 \pmod 9$ and $\text{step}(c) = 1$. If $y \equiv 2$, then $E_c \equiv 1$ and $E_d \equiv 0$, so $c \equiv 8 \pmod 9$ and $\text{step}(c) = 2$. In both cases $\text{step}(c) \equiv y \pmod 3$, hence $E(n, c) = y$.

By Lemma 5.2, $\rho^n(c) \equiv c \cdot 4^y \equiv 2 \cdot 4^{E_c+y} \equiv 2 \cdot 4^{E_d+3^K E_c} \equiv 2 \cdot 4^{E_d}(4^{3^K})^{E_c} \pmod{3^{K+1}}$. By Lemma 5.4, $4^{3^K} \equiv 1 \pmod{3^{K+1}}$, so $\rho^n(c) \equiv 2 \cdot 4^{E_d} \equiv d \pmod{3^{K+1}}$. □

Proposition 5.8 (Full Loop at 8)**.** *For every $K \geq 1$, there exists $n > 0$ such that $\rho^n(8) = 8$ and $E(n, 8) = 3^K$.*

Proof. Write $K = j + 1$. Choose $n = 2 \cdot 3^j$. By Lemma 5.3, $E(n, 8) = 3 \cdot 3^j = 3^{j+1} = 3^K$. Then by Lemma 5.2, $\rho^n(8) \equiv 8 \cdot 4^{3^K} \equiv 8 \cdot 1 = 8 \pmod{3^{K+1}}$ using Lemma 5.4. Since $8 < 3^{K+1}$ (as $K \geq 1$), $\rho^n(8) = 8$. □

6. Affine Labels and Holonomy

Throughout Sections 6 through 8, we assume $p \geq 5$. The non-zero elements $2, 3, 4, 6 \in \mathbb{Z}/p\mathbb{Z}$ are therefore invertible.

Definition 6.1 (Affine Group)**.** Define $\text{Aff}(p) = \mathbb{Z}/p\mathbb{Z}^\times \times \mathbb{Z}/p\mathbb{Z}$. An element $f = (f_1, f_2) \in \text{Aff}(p)$ acts on $\mathbb{Z}/p\mathbb{Z}$ by $f(t) = f_1 t + f_2$. Composition in diagrammatic order (first f, then g) is defined by:

$$f \,;\, g = (g_1 f_1,\ g_1 f_2 + g_2),$$

which satisfies $(f \,;\, g)(t) = g(f(t))$. The identity is $(1, 0)$, and the inverse is:

$$f^{-1} = (f_1^{-1},\ -f_1^{-1} f_2).$$

Note that $\text{Aff}(p)$ is strictly non-abelian, and $(f \,;\, g)^{-1} = g^{-1} \,;\, f^{-1}$.

Definition 6.2 (Edge and Walk Labels)**.** Fix depth K. The class graph has vertices given by valid classes modulo 3^{K+1}.

- **Right edges:** For each valid class $c < 3^{K+1}$, an edge $c \to \rho(c)$ with label

$$\text{rl}(c) = (4^{\text{step}(c)},\ 0) \in \text{Aff}(p).$$

- **Left edges:** For each valid $e < 3^{K+2}$, an edge $(e \bmod 3^{K+1}) \to \lambda(e)$ with label

$$\text{ll}(e) = \left(\frac{\mu(e)}{3},\ -\frac{\mu(e)}{6}\right) \in \text{Aff}(p).$$

A *walk* from u to w is a finite alternating sequence of forward and backward edge traversals. The label of a walk is the composition of its edge labels: forward traversals compose with f, backward traversals with f^{-1}. The label of the empty walk at c is $(1, 0)$.

Definition 6.3 (Holonomy Group)**.** The holonomy group at basepoint 8 at depth K is:

$$\text{Hol}(K) = \{f \in \text{Aff}(p) : \text{there exists a closed walk from 8 to 8 with label } f\}.$$

By construction, $\text{Hol}(K)$ is a subgroup of $\text{Aff}(p)$.

Lemma 6.4 (Operator Equivariance Modulo p)**.** *Let $x \in G$.*

(1) $(R(x) : \mathbb{Z}/p\mathbb{Z}) = \text{rl}(x)(x : \mathbb{Z}/p\mathbb{Z})$.

(2) $(L(x) : \mathbb{Z}/p\mathbb{Z}) = \text{ll}(x)(x : \mathbb{Z}/p\mathbb{Z})$.

Proof. (1) Immediate from $R(x) = x \cdot 4^{\text{step}(x)}$.

(2) By Lemma 4.1, $6L(x) = \mu(x)(2x - 1)$ holds in $\mathbb{N}$. Reducing modulo p:

$$6(L(x) : \mathbb{Z}/p\mathbb{Z}) = \mu(x)(2(x : \mathbb{Z}/p\mathbb{Z}) - 1).$$

On the other hand,

$$6 \cdot \mathrm{ll}(x)(x : \mathbb{Z}/p\mathbb{Z}) = 6\left(\frac{\mu(x)}{3}(x : \mathbb{Z}/p\mathbb{Z}) - \frac{\mu(x)}{6}\right) = 2\,\mu(x)(x : \mathbb{Z}/p\mathbb{Z}) - \mu(x).$$

Since $6 \neq 0$ in $\mathbb{Z}/p\mathbb{Z}$, we cancel 6 to obtain the result. □

Lemma 6.5 (Walks from Right Iterates)**.** *Let $c < 3^{K+1}$ be valid. For any $n \in \mathbb{N}$, there is a forward right-walk from c to $\rho^n(c)$ with label:*

$$(4^{E(n,c)},\ 0) \in \mathrm{Aff}(p).$$

Proof. Induction on n. At $n = 0$, the label is $(4^0, 0) = (1, 0)$. At $n + 1$, prepending the edge $c \to \rho(c)$ with label $(4^{\mathrm{step}(c)}, 0)$ composes to $(4^{E(n,\rho(c))} \cdot 4^{\mathrm{step}(c)}, 0) = (4^{E(n+1,c)}, 0)$. □

7. Fibre Stabilization and Exact Label Action

Fix a seed $v \in G$.

Definition 7.1 (Adic Fibres)**.** For $K \geq 1$ and a valid class $c < 3^{K+1}$, the fibre of c is:

$$F_K(c) = \{(n : \mathbb{Z}/p\mathbb{Z}) \in \mathbb{Z}/p\mathbb{Z} : n \in \mathcal{D}(v) \text{ and } n \equiv c \pmod{3^{K+1}}\}.$$

Lemma 7.2 (Fibre Dynamics)**.** *(1) For any valid class $c < 3^{K+1}$, $\mathrm{rl}(c)(F_K(c)) \subseteq F_K(\rho(c))$.*

(2) For any valid $e < 3^{K+2}$, $\mathrm{ll}(e)(F_{K+1}(e)) \subseteq F_K(\lambda(e))$.

(3) If $J \leq K$ and $c < 3^{K+1}$ is valid, then $F_K(c) \subseteq F_J(c \bmod 3^{J+1})$.

(4) $F_K(v \bmod 3^{K+1})$ is non-empty for all $K \geq 1$ (witnessed by $v \in \mathcal{D}(v)$).

Proof. (1) If $t \in F_K(c)$, $t = (n : \mathbb{Z}/p\mathbb{Z})$ for some $n \in \mathcal{D}(v)$ with $n \equiv c \pmod{3^{K+1}}$. Then $R(n) \in \mathcal{D}(v)$, $R(n) \bmod 3^{K+1} = \rho(c)$ (by reduction), and $(R(n) : \mathbb{Z}/p\mathbb{Z}) = \mathrm{rl}(c)(t)$ by Lemma 6.4(1).

(2) Similar: $L(n) \in \mathcal{D}(v)$, Lemma 4.4 gives $L(n) \equiv \lambda(e) \pmod{3^{K+1}}$, and Lemma 6.4(2) matches the label.

(3) Follows from $3^{J+1} \mid 3^{K+1}$. (4) $v \in \mathcal{D}(v)$ gives $(v : \mathbb{Z}/p\mathbb{Z}) \in F_K(v \bmod 3^{K+1})$. □

Proposition 7.3 (Cardinality Stabilization)**.** *There exist $K_0 \geq 1$ and $\sigma \in \mathbb{N}$ with $\sigma \geq 1$ such that for all $K \geq K_0$ and all valid classes $c < 3^{K+1}$:*

$$|F_K(c)| = \sigma.$$

Proof. Since right-edge labels have non-zero slopes $4^{\mathrm{step}(c)} \in \mathbb{Z}/p\mathbb{Z}^\times$, they act injectively. By Lemma 7.2(1), $|F_K(c)| \leq |F_K(\rho(c))|$. Applying transitivity (Proposition 5.7) in both directions yields:

$$|F_K(c)| = |F_K(d)|$$

for all valid classes $c, d < 3^{K+1}$. In particular, $|F_K(c)| = |F_K(8)|$.

Next, by Lemma 7.2(3) at $J = K$ and class 8, $F_{K+1}(8) \subseteq F_K(8)$, so the sequence of natural numbers $k \mapsto |F_k(8)|$ is non-increasing. By Lemma 7.2(4) and transitivity, $|F_K(8)| = |F_K(v \bmod 3^{K+1})| \geq 1$ for all K. A non-increasing sequence of positive integers must stabilize: there exist $K_0 \geq 1$ and $\sigma \geq 1$ such that $|F_K(8)| = \sigma$ for all $K \geq K_0$. □

Lemma 7.4 (Fibre Equality Across Successive Depths)**.** *Let $K \geq K_0$. For any valid $e < 3^{K+2}$:*

$$F_{K+1}(e) = F_K(e \bmod 3^{K+1}).$$

Proof. By Lemma 7.2(3), $F_{K+1}(e) \subseteq F_K(e \bmod 3^{K+1})$. By Proposition 7.3, both sets are finite and have identical cardinality σ. A subset of a finite set with equal cardinality is the entire set. □

Proposition 7.5 (Exact Label Action)**.** *Let $K \geq K_0$. For any edge $u \to w$ with label f, $f(F_K(u)) = F_K(w)$. Consequently, for any walk from u to w with label g:*

$$g(F_K(u)) = F_K(w).$$

In particular, every holonomy element $h \in \mathrm{Hol}(K)$ acts as a bijection on $F_K(8)$.

Proof. For a right edge $c \to \rho(c)$, $\mathrm{rl}(c)$ is injective and maps $F_K(c)$ into $F_K(\rho(c))$. Since $|F_K(c)| = \sigma = |F_K(\rho(c))|$, the image is the entire target. For a left edge $(e \bmod 3^{K+1}) \to \lambda(e)$, Lemma 7.4 gives $F_{K+1}(e) = F_K(e \bmod 3^{K+1})$. Thus $\mathrm{ll}(e)$ maps $F_K(e \bmod 3^{K+1})$ into $F_K(\lambda(e))$, which again is surjective by cardinality comparison. The result for general walks follows by induction on the walk length. □

8. The Commutator Engine and Simultaneous Density

Lemma 8.1 (Commutator Identity)**.** *Let $\lambda, \alpha \in \mathbb{Z}/p\mathbb{Z}^\times$ and $\beta \in \mathbb{Z}/p\mathbb{Z}$. Then:*

$$(\alpha, \beta)^{-1} \,;\, (\lambda, 0)^{-1} \,;\, (\alpha, \beta) \,;\, (\lambda, 0) = (1,\ (\lambda - 1)\beta).$$

Proof. We evaluate the composition stepwise according to Definition 6.1:

$$(\alpha, \beta)^{-1} = (\alpha^{-1}, -\alpha^{-1}\beta),$$
$$(\lambda, 0)^{-1} = (\lambda^{-1}, 0).$$

Composing the first two factors:

$$(\alpha^{-1}, -\alpha^{-1}\beta) \,;\, (\lambda^{-1}, 0) = (\lambda^{-1}\alpha^{-1}, -\lambda^{-1}\alpha^{-1}\beta).$$

Composing with (α, β):

$$\begin{aligned}(\lambda^{-1}\alpha^{-1}, -\lambda^{-1}\alpha^{-1}\beta) \,;\, (\alpha, \beta) &= (\alpha\lambda^{-1}\alpha^{-1},\ \alpha(-\lambda^{-1}\alpha^{-1}\beta) + \beta) \\ &= (\lambda^{-1},\ (1 - \lambda^{-1})\beta).\end{aligned}$$

Finally, composing with $(\lambda, 0)$:

$$(\lambda^{-1},\ (1 - \lambda^{-1})\beta) \,;\, (\lambda, 0) = (\lambda\lambda^{-1},\ \lambda(1 - \lambda^{-1})\beta + 0) = (1,\ (\lambda - 1)\beta).$$ □

Lemma 8.2 (Walk with Non-Zero Translation)**.** *For every $K \geq 1$, there exist $\alpha, \beta \in \mathbb{Z}/p\mathbb{Z}$ with $\alpha \neq 0$ and $\beta \neq 0$ such that $(\alpha, \beta) \in \mathrm{Hol}(K)$.*

Proof. Consider $e = 8$. Since $8 < 3^{K+2}$, there is a left edge from $8 \bmod 3^{K+1} = 8$ to $\lambda(8) = L(8) \bmod 3^{K+1}$ with label $\mathrm{ll}(8)$. By Proposition 5.7, choose a right-path from $\lambda(8)$ back to 8; by Lemma 6.5, this path has label $(4^E, 0)$ for some $E \in \mathbb{N}$. Composing these yields a closed walk at 8 with label:

$$(\alpha, \beta) = \mathrm{ll}(8) \,;\, (4^E, 0) = \left(4^E \frac{\mu(8)}{3},\ -4^E \frac{\mu(8)}{6}\right).$$

Since $p \geq 5$, $4 \neq 0$, and $\mu(8) = 8 \neq 0$, the second coordinate is $\beta = -4^E(8/6) \neq 0$. □

Lemma 8.3 (Pure Dilation from Loop Pairs)**.** *Let $e_1, e_2 < 3^{K+2}$ be valid classes. Suppose there exist right-paths:*

- *$u_1 \to u_2$ with $4^{E(n_s,u_1)} = 1$ in $\mathbb{Z}/p\mathbb{Z}$, where $u_i = e_i \bmod 3^{K+1}$,*
- *$w_1 \to w_2$ where $w_i = \lambda(e_i)$, with cumulative exponent $E_t = E(n_t, w_1)$.*

Then:

$$\left(\frac{\mu(e_2)}{\mu(e_1)\cdot 4^{E_t}},\ 0\right) \in \mathrm{Hol}(K).$$

Proof. Let W_s be the walk $u_1 \to u_2$ with label $(1,0)$, and W_t the walk $w_1 \to w_2$ with label $(4^{E_t}, 0)$. Let E_1 be the left edge $u_1 \to w_1$ (label $\mathrm{ll}(e_1)$) and E_2 the left edge $u_2 \to w_2$ (label $\mathrm{ll}(e_2)$). Form the closed loop based at w_1:

$$\Omega = E_1^{-1}\,;W_s\,;E_2\,;W_t^{-1}.$$

Evaluating the label of Ω:

$$\mathrm{ll}(e_1)^{-1} = \left(\frac{3}{\mu(e_1)},\ \frac{1}{2}\right).$$

Composing with W_s (which acts as the identity $(1,0)$) leaves the label unchanged. Composing with $\mathrm{ll}(e_2) = (\mu(e_2)/3, -\mu(e_2)/6)$:

$$\left(\frac{3}{\mu(e_1)},\ \frac{1}{2}\right);\left(\frac{\mu(e_2)}{3},\ -\frac{\mu(e_2)}{6}\right) = \left(\frac{\mu(e_2)}{\mu(e_1)},\ \frac{\mu(e_2)}{3}\left(\frac{1}{2}\right) - \frac{\mu(e_2)}{6}\right) = \left(\frac{\mu(e_2)}{\mu(e_1)},\ 0\right).$$

Finally, composing with $W_t^{-1} = (4^{-E_t}, 0)$ gives the label $(\mu(e_2)/(\mu(e_1)4^{E_t}), 0)$. Connecting 8 to w_1 by a right-path γ and conjugating Ω by γ yields a closed walk based at 8 with identical label. □

Lemma 8.4 (Existence of Non-Trivial Dilations)**.** *For every $K \geq 1$, there exists $\lambda \in \mathbb{Z}/p\mathbb{Z}^\times$ with $\lambda \neq 1$ such that $(\lambda, 0) \in \mathrm{Hol}(K)$.*

Proof. We divide the analysis into three exhaustive cases:

Case 1: $4^{3^K} \neq 1$ in $\mathbb{Z}/p\mathbb{Z}$. By Proposition 5.8, there is a closed walk at 8 with cumulative exponent $E = 3^K$. By Lemma 6.5, its label is $(4^{3^K}, 0) \in \mathrm{Hol}(K)$. Take $\lambda = 4^{3^K} \neq 1$.

Case 2: $4^{3^K} = 1$ in $\mathbb{Z}/p\mathbb{Z}$ and $\mathrm{ord}(4) = 3^a$ with $2 \leq a \leq K$. Write $a = j + 2$. Set:

$$e_1 = 8, \qquad e_2 = 8 + 27 \cdot 3^j = 8 + 3^{a+1}.$$

Both e_1, e_2 are valid and bounded by 3^{K+2}. By Lemma 4.2, $\mu(e_1) = \mu(e_2) = 8$. By Lemma 4.1, $L(e_1) = 20$ and $L(e_2) = 20 + 72 \cdot 3^j$. Let $u_i = e_i \bmod 3^{K+1}$ and $w_i = \lambda(e_i)$.

- *Source path:* By Proposition 5.7, take a path $u_1 \to u_2$ with exponent E_s. Since $e_2 \equiv e_1 \pmod{3^{a+1}}$, Lemma 5.2 forces $4^{E_s} \equiv 1 \pmod{3^{a+1}}$, so $3^a \mid E_s$ by Lemma 5.4. Since $\mathrm{ord}(4) = 3^a$, $4^{E_s} = 1$ in $\mathbb{Z}/p\mathbb{Z}$.
- *Target path:* Let $w_1 \to w_2$ have exponent E_t. If $4^{E_t} = 1$ in $\mathbb{Z}/p\mathbb{Z}$, then $3^a \mid E_t$, which would force $L(e_2) \equiv L(e_1) \pmod{3^{a+1}}$, meaning $72 \cdot 3^j \equiv 0 \pmod{27 \cdot 3^j}$, or $27 \mid 72$, a contradiction. Thus $4^{E_t} \neq 1$ in $\mathbb{Z}/p\mathbb{Z}$.

Applying Lemma 8.3 yields $\lambda = 8/(8 \cdot 4^{E_t}) = 4^{-E_t} \neq 1$ in $\mathbb{Z}/p\mathbb{Z}$.

Case 3: $4^{3^K} = 1$ in $\mathbb{Z}/p\mathbb{Z}$ and $\mathrm{ord}(4) = 3$. Take $e_1 = 2$ and $e_2 = 11$. Both are valid and $< 3^{K+2}$. Here $\mu(2) = 4$, $\mu(11) = 16$, $L(2) = 2$, and $L(11) = 56$. Notice that:

$$u_1 \equiv 2, \quad u_2 \equiv 2, \quad w_1 = 2 \equiv 2, \quad w_2 = 56 \equiv 2 \pmod 9.$$

Because both endpoints of the source path $u_1 \to u_2$ and target path $w_1 \to w_2$ are $\equiv 2 \pmod 9$, Lemma 5.2 reduced modulo 9 gives $2 \equiv 2 \cdot 4^E \pmod 9$ for $E \in \{E_s, E_t\}$, and $4^E \bmod 9$ is $1, 4, 7$

according as $E \bmod 3$ is $0, 1, 2$; hence $3 \mid E_s$ and $3 \mid E_t$. Since $\mathrm{ord}(4) = 3$, $4^{E_s} = 4^{E_t} = 1$ in $\mathbb{Z}/p\mathbb{Z}$. Lemma 8.3 yields:

$$\lambda = \frac{\mu(11)}{\mu(2) \cdot 1} = \frac{16}{4} = 4 \in \mathbb{Z}/p\mathbb{Z}^{\times}.$$

Here $\lambda \neq 0$ because $4 \neq 0$ in $\mathbb{Z}/p\mathbb{Z}$. If $\lambda = 1$, then $16 = 4$, so $12 = 0$ in $\mathbb{Z}/p\mathbb{Z}$, i.e. $3 \cdot 4 = 0$. Since $p \geq 5$, both $3 \neq 0$ and $4 \neq 0$, contradicting the absence of zero divisors in $\mathbb{Z}/p\mathbb{Z}$. Hence $\lambda \neq 1$.

The cases are exhaustive: either $4^{3^K} \neq 1$ in $\mathbb{Z}/p\mathbb{Z}$ (Case 1), or $4^{3^K} = 1$, which forces $\mathrm{ord}(4) \mid 3^K$. In the latter regime, since $p \geq 5$ precludes $\mathrm{ord}(4) = 1$ (as $4 \equiv 1 \pmod{p} \implies p \mid 3$), we must have $\mathrm{ord}(4) = 3^a$ for some $1 \leq a \leq K$, splitting into $a = 1$ (Case 3) and $2 \leq a \leq K$ (Case 2). □

Proposition 8.5 (Existence of Translations in Holonomy)**.** *For every $K \geq 1$, there exists $d \in \mathbb{Z}/p\mathbb{Z}$ with $d \neq 0$ such that $(1, d) \in \mathrm{Hol}(K)$.*

Proof. Choose $(\alpha, \beta) \in \mathrm{Hol}(K)$ with $\alpha, \beta \neq 0$ by Lemma 8.2. Choose $(\lambda, 0) \in \mathrm{Hol}(K)$ with $\lambda \neq 0, 1$ by Lemma 8.4. Since $\mathrm{Hol}(K)$ is a group, it contains the commutator:

$$(\alpha, \beta)^{-1} ; (\lambda, 0)^{-1} ; (\alpha, \beta) ; (\lambda, 0).$$

By Lemma 8.1, this element equals $(1, d)$ with $d = (\lambda - 1)\beta$. Since $\mathbb{Z}/p\mathbb{Z}$ is a field and $\lambda \neq 1$, $\beta \neq 0$, we have $d \neq 0$. □

Theorem 8.6 (Simultaneous Density)**.** *Let $p \geq 5$ and $v \in G$. For any $J \geq 1$, any valid class $c < 3^{J+1}$, and any residue $r \in \mathbb{Z}/p\mathbb{Z}$, there exists $y \in \mathcal{D}(v)$ such that:*

$$y \equiv c \pmod{3^{J+1}} \quad \textit{and} \quad (y : \mathbb{Z}/p\mathbb{Z}) = r.$$

Proof. Let K_0, σ be the stabilization parameters from Proposition 7.3. Fix $K = \max(K_0, J)$.

By Proposition 8.5, there exists a non-trivial translation $(1, d) \in \mathrm{Hol}(K)$ with $d \neq 0$. By Proposition 7.5, $(1, d)$ maps $F_K(8)$ onto itself:

$$t \in F_K(8) \implies t + d \in F_K(8).$$

Since $F_K(8)$ is non-empty, choose $t_0 \in F_K(8)$. By induction, $t_0 + kd \in F_K(8)$ for all $k \in \mathbb{N}$. Because $d \in \mathbb{Z}/p\mathbb{Z}^{\times}$, the additive orbit $\{t_0 + kd : k \in \mathbb{N}\}$ equals all of $\mathbb{Z}/p\mathbb{Z}$. Thus $F_K(8) = \mathbb{Z}/p\mathbb{Z}$.

This implies that the stabilized cardinality is $\sigma = |F_K(8)| = p$. Since $c < 3^{J+1} \leq 3^{K+1}$, c is a valid class at depth K, so $|F_K(c)| = \sigma = p$, which forces $F_K(c) = \mathbb{Z}/p\mathbb{Z}$. Thus $r \in F_K(c)$.

Since $J \leq K$, Lemma 7.2(3) gives $r \in F_J(c \bmod 3^{J+1}) = F_J(c)$. By Definition 7.1, there exists $y \in \mathcal{D}(v)$ satisfying $y \equiv c \pmod{3^{J+1}}$ and $(y : \mathbb{Z}/p\mathbb{Z}) = r$. □

9. Small Primes and the Master Theorem

Proposition 9.1 (Case $p \geq 5$)**.** *Theorem 1.1 holds for $p \geq 5$.*

Proof. By Lemma 3.6, choose $v \in G$ reaching w. Let $J = B+1$ and define $c = 9 \cdot 3^B - 1 = 3^{B+2} - 1$. Then $c \equiv 8 \pmod{9}$, so c is valid; $c < 3^{J+1}$; and $c > B$ (since $B < 3^B \leq c$). Applying Theorem 8.6 with class c and residue $r \in \mathbb{Z}/p\mathbb{Z}$ yields $y \in \mathcal{D}(v)$ such that $y \equiv c \pmod{3^{J+1}}$ and $(y : \mathbb{Z}/p\mathbb{Z}) = r$.

Since $y \bmod 3^{J+1} = c$ and $c > 0$, we have $y \geq c > B$. By Lemma 3.5, $y \rightsquigarrow v$, and since $v \rightsquigarrow w$, transitivity ensures $y \rightsquigarrow w$. □

Proposition 9.2 (Case $p = 2$)**.** *Theorem 1.1 holds for $p = 2$.*

Proof. Let $r \in \mathbb{Z}/2\mathbb{Z}$.

- If $r = 0$: set $n = 2^{B+1}w$. Then $n > B$, $n \equiv 0 \pmod{2}$, and $n \rightsquigarrow w$ in $B + 1$ halvings.

- If $r = 1$: choose $t = 3B + 3$. There exists $k \in \{2t, 2t + 1\}$ such that $2^k w \equiv 1 \pmod 3$ (if $w \equiv 1 \pmod 3$, choose $k = 2t$; if $w \equiv 2 \pmod 3$, choose $k = 2t + 1$). Since $k \geq 6B + 6$, $2^k w > 6B + 6$. Define $n = (2^k w - 1)/3$. Then $3n + 1 = 2^k w$. Since $2^k w$ is even, n is odd, so $(n : \mathbb{Z}/2\mathbb{Z}) = 1$. Furthermore, $n > B$, and $\mathcal{C}(n) = 2^k w$, which reaches w in k halvings. □

Proposition 9.3 (Case $p = 3$)**.** *Theorem 1.1 holds for $p = 3$.*

Proof. Let $r \in \mathbb{Z}/3\mathbb{Z}$.

- If $r = 0$: 2 is a primitive root modulo 9. There exists $j < 6$ such that $2^j w \equiv 1 \pmod 9$. Set $k = j + 6(B + 1)$. Since $64 \equiv 1 \pmod 9$, $2^k w = 64^{B+1} \cdot 2^j w \equiv 1 \pmod 9$. Set $n = (2^k w - 1)/3$. Since $2^k w$ is even, n is odd, so $\mathcal{C}(n) = 3n + 1 = 2^k w$, and $n \rightsquigarrow w$. Since $2^k w \equiv 1 \pmod 9$, $n \equiv 0 \pmod 3$. Finally, $n > B$.
- If $r = 1$: choose $k \geq B$ such that $2^k w \equiv 1 \pmod 3$. Set $n = 2^k w$. Then $n > B$, $n \equiv 1 \pmod 3$, and $n \rightsquigarrow w$ in k halvings.
- If $r = 2$: with k as above, set $n = 2^{k+1} w$. Then $n \equiv 2 \cdot 1 = 2 \pmod 3$, $n > B$, and $n \rightsquigarrow w$. □

Proof of Theorem 1.1. Combine Propositions 9.1, 9.2, and 9.3. □

10. Application: Local Obstruction Barriers for Cycles

A hypothetical Collatz cycle with k odd steps and S total steps satisfies the Diophantine equation:

$$(2^S - 3^k)n = N,$$

where $N = \sum_{i=0}^{k-1} 3^{k-1-i} 2^{s_i}$ is a positive integer determined by the step sequence.

Definition 10.1 (Cycle Specification)**.** A *cycle specification* is a triple $(k, S, N) \in \mathbb{N}^3$.

Theorem 10.2 (Predecessors Modulo p)**.** *Let p be any prime. If $v > 0$ with $3 \nmid v$, then for every $r \in \mathbb{Z}/p\mathbb{Z}$, there exists $n \in \mathbb{N}$ such that $n \rightsquigarrow v$ and $(n : \mathbb{Z}/p\mathbb{Z}) = r$.*

Proof. Specialization of Theorem 1.1 with threshold $B = 0$. □

Theorem 10.3 (Absence of Local Obstructions)**.** *Let p be any prime. Let (k, S, N) be a cycle specification such that*

$$(2^S - 3^k : \mathbb{Z}/p\mathbb{Z}) \neq 0.$$

Then for every target $v > 0$ with $3 \nmid v$, there exists $n \in \mathbb{N}$ such that $n \rightsquigarrow v$ and

$$(2^S - 3^k : \mathbb{Z}/p\mathbb{Z}) \cdot (n : \mathbb{Z}/p\mathbb{Z}) = (N : \mathbb{Z}/p\mathbb{Z}).$$

Proof. Since $(2^S - 3^k : \mathbb{Z}/p\mathbb{Z}) \neq 0$ and $\mathbb{Z}/p\mathbb{Z}$ is a field, the linear equation has a unique solution:

$$r = (2^S - 3^k : \mathbb{Z}/p\mathbb{Z})^{-1} (N : \mathbb{Z}/p\mathbb{Z}) \in \mathbb{Z}/p\mathbb{Z}.$$

By Theorem 10.2, there exists $n \in \mathbb{N}$ reaching v with $(n : \mathbb{Z}/p\mathbb{Z}) = r$. Multiplying by $(2^S - 3^k : \mathbb{Z}/p\mathbb{Z})$ gives $(2^S - 3^k : \mathbb{Z}/p\mathbb{Z}) \cdot (n : \mathbb{Z}/p\mathbb{Z}) = (N : \mathbb{Z}/p\mathbb{Z})$. □

Remark 10.4. Theorem 10.3 demonstrates that for any prime p where the cycle denominator does not vanish, modular conditions modulo p alone cannot rule out the existence of cycle predecessors. The global obstruction to non-trivial cycles must necessarily involve multi-prime interactions or archimedean constraints.

Formalization Correspondence Table

Table 1. Correspondence between paper mathematical statements and verified declarations in `ERE.lean`.

Paper Definition / Result	**Lean 4 Formalization (`ERE.*`)**
Collatz operator $\mathcal{C}$, reachability $\rightsquigarrow$	`collatz`, `Reaches`
Lemma 1.2 (Halving, atomic inverse steps)	`collatz_double`, `collatz_pow_mul`, `reaches_trans`, `collatz_eq_iff`
Corollary 1.3 (Multiples of 3)	`pred_of_three_dvd`
Good numbers, coordinate $x(n)$	`GoodOdd`, `toX`
Lemma 2.2 (Coordinate step)	`collatz_odd`, `reaches_toX`
Valid class, domain G	`ValidClass`, `InG`
Lemma 2.5 (Good numbers are valid classes)	`toX_inG`, `exists_goodOdd_of_inG`, `inG_iff`
Branches R, L, multiplier μ	`R`, `L`, `mult`
Abstract descendant tree $\mathcal{D}(v)$	`Desc`
Proposition 3.3 (Derivation of branches)	`R_first`, `L_first`, `collatz_third_pred`, `Rorig`, `multOrig`, `Lorig`, `R_conj`, `L_conj`
Lemma 3.4 (Branch trajectories)	`R_reaches`, `L_reaches`
Lemma 3.5 (Soundness of $\mathcal{D}$)	`Desc_reaches`
Lemma 3.6 (Seed existence)	`exists_inG_reaches`
Lemma 4.1 (Exactness of L)	`L_spec`
Lemma 4.2 (Residues mod 27)	`residue_mod27`, `mult_mem`, `mult_pow`
Lemma 4.3 (Invariance of G)	`R_mem`, `L_mem`, `L_valid`, `Desc_mem`
Lemma 4.4 (Precision loss under L)	`mod27_eq_of_mod_pow`, `mult_congr`, `L_cong`
$\rho, \lambda, \text{step}, \rho^n, E(n,c)$	`rnext`, `lnext`, `rStep`, `rIter`, `rExp`
Lemma 5.2 (Multiplication action)	`rIter_eq`, `rIter_eq'`
Lemma 5.3 (Two-step tripling)	`rStep_pair`, `rExp_two_mul`
Lemma 5.4 (LTE for base 4)	`lte_four`, `four_pow_modEq_one_iff`
Lemma 5.5 (Four-cancellation)	`four_cancel`
Lemma 5.6 (Exponent classification)	`four_pow_mod9`, `exp_type`, `exists_exp`
Proposition 5.7 (Transitivity of ρ)	`R_transitive`
Proposition 5.8 (Loop at 8)	`R_full_loop`
Affine group $\text{Aff}(p)$, holonomy $\text{Hol}(K)$	`Aff`, `aapp`, `athen`, `ainv`, `Edge`, `Walk`, `Hol`
Lemma 6.4 (Operator equivariance)	`R_label`, `L_label`, `rightLabel_fst`, `rightLabel_snd`
Lemma 6.5 (Walks from right iterates)	`walk_of_rIter`
Fibres $F_K(c)$	`M`, `fibre`
Lemma 7.2 (Fibre dynamics)	`fibre_right`, `fibre_left`, `fibre_proj`, `fibre_root_nonempty`
Proposition 7.3 (Fibre stabilization)	`fibre_ncard_step`, `fibre_ncard_chain`, `fibre_ncard_const`, `stabilization`
Lemma 7.4 (Successive depths)	`fibre_succ_eq`
Proposition 7.5 (Exact label action)	`edge_image`, `walk_image`
Lemma 8.1 (Commutator identity)	`comm_label`
Lemma 8.2 (Reference walk)	`ref_walk`
Lemma 8.3 (Pair dilation)	`rpath`, `pure_dilation_of_pair`
Lemma 8.4 (Dilation existence)	`dil_case1`, `dil_case2a`, `dil_case2b`, `order_split`, `dilation_exists`
Proposition 8.5 (Translation in Hol)	`translation_of_dilation`, `translation_exists`
Theorem 8.6 (Simultaneous density)	`main_theorem_core`
Proposition 9.1 ($p \geq 5$)	`main_theorem_p_ge_5`
Proposition 9.2 ($p = 2$)	`main_theorem_p_eq_2`
Proposition 9.3 ($p = 3$)	`main_theorem_p_eq_3`
Theorem 1.1 (Master Theorem)	`main_theorem_infinite`
Cycle Specification	`CycleSpec`
Theorem 10.2 (Reaching density)	`reaching_predecessor_dense_mod_p`
Theorem 10.3 (Cycle barrier)	`cycle_equation_solvable_mod_p`, `cycle_local_obstruction_barrier`

TABLE 2. Auxiliary Lean 4 declarations supporting the primary equivalence.

Auxiliary Facts	**Lean 4 Declarations (`ERE.*`)**
Classes, exponents	`nine_dvd_pow`, `validClass_mod`, `valid8`, `eight_lt`, `rnext_valid`, `rIter_valid`, `rIter_modEq`, `exp_dvd_of_congr`, `congr_of_exp_dvd`, `rExp_mod3`, `four_pow_eq_one_of_mod3`
Affine labels, walks, fibres	`athen_app`, `athen_assoc`, `athen_one_left`, `athen_one_right`, `ainv_app`, `ainv_ainv`, `ainv_athen`, `aapp_inj`, `two_ne_zero_zmod`, `three_ne_zero_zmod`, `four_ne_zero_zmod`, `six_ne_zero_zmod`, `mult_cast_ne_zero`, `Edge.label_unit`, `Walk.label_unit`, `Walk.append`, `Walk.symm`, `Hol_mul`, `Hol_inv`, `HasTranslation`, `R_class`, `L_class`, `rightLabel_mod`, `leftLabel_mod`, `fibre_ncard_antitone`
Collatz bridge, small primes	`collatz_8`, `inverse_tree_covers`, `exists_large_valid_class`, `cast_eq_of_mod`, `cast_eq_of_mod'`, `exists_pow_two_mod`, `exists_k_mod_three`, `four_pow_mod_three`, `sixtyfour_pow_mod`, `lt_pow_mul`, `odd_of_three_mul_add_one`, `reaches_of_three_mul_add_one`, `ge_of_mod_eq`

Formalization Architecture and Verification Environment. The complete formalization is self-contained in `ERE.lean`. The development introduces zero custom axioms and contains no unproven assertions (`sorry`). Running Lean's `#print axioms main_theorem_infinite` confirms that the verification rests strictly upon standard classical foundations: `propext`, `Classical.choice`, and `Quot.sound`. The project builds under Lean 4 toolchain `leanprover/lean4:v4.36.0-rc1`, Mathlib commit `0b57f69`, Batteries `8e746d7` and Aesop `29c64c5`. The repository, including `lakefile.toml` and `lake-manifest.json`, is available at

https://github.com/Sio-MLST/Every-Remainder-Everywhere